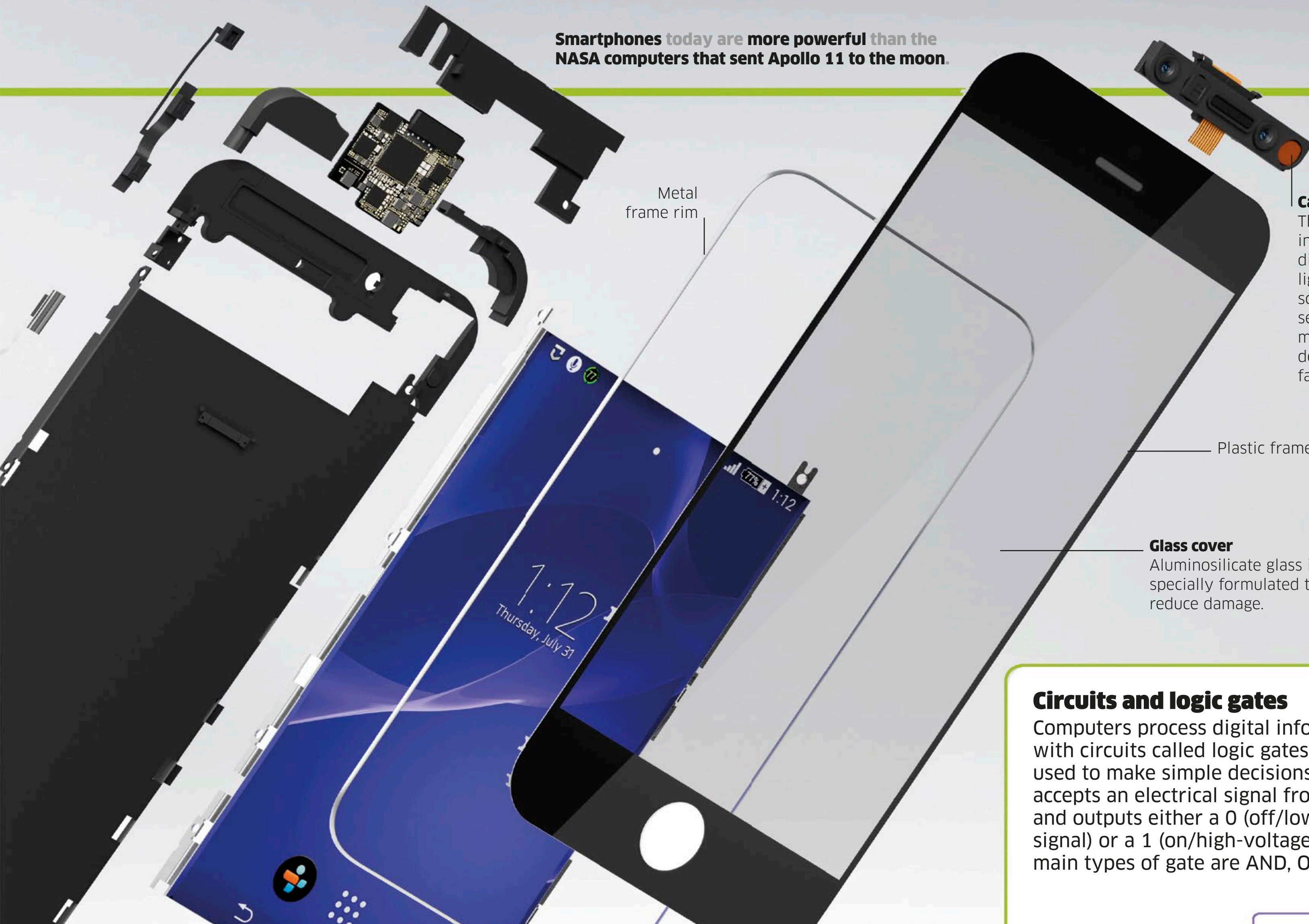
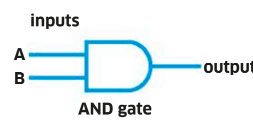


Circuits and logic gates

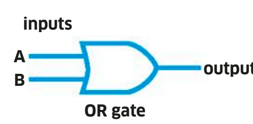
Computers process digital information with circuits called logic gates, which are used to make simple decisions. A logic gate accepts an electrical signal from its inputs and outputs either a 0 (off/low-voltage signal) or a 1 (on/high-voltage signal). The main types of gate are AND, OR, and NOT.



AND gate

Input A	Input B	Output
1	0	0
0	1	0
0	0	0
1	1	1

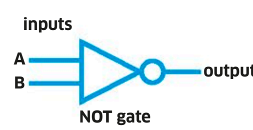
AND gate
This compares the two numbers and switches on only if both the numbers are 1. There will only be an output if both inputs are on.



OR gate

Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

OR gate
This switches on if either of the two numbers is 1. If both numbers are 0, it switches off. There will be an output if one or both inputs are on.



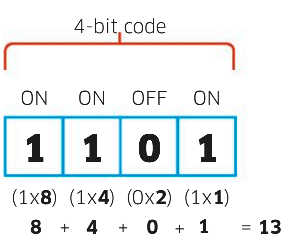
NOT gate

Input	Output
0	1
1	0

NOT gate
This reverses (inverts) whatever goes into it. A 0 becomes a 1, and vice versa. The output is only on if the input is off. If the input is on, the output is off.

Digital electronics

Most technology we use today is digital. Our devices convert information into numbers or digits and process these numbers in place of the original information. Digital cameras turn images into patterns of numbers, while cell phones send and receive calls with signals representing strings of numbers. These are sent in a code called binary, using only the numerals 1 and 0 (rather than decimal, 0–9).



Binary numbers
In binary, the position of 1s and 0s corresponds to a decimal value. Each binary position doubles in decimal value from right to left (1, 2, 4, 8) and these values are either turned on (x1) or off (x0). In the 4-bit code shown, the values of 8, 4, and 1 are all "on," and when added together equal 13.

Analog to digital

A sound wave made by a musical instrument is known as analog information. The wave rises and falls as the sound rises and falls. A wave can be measured at different points to produce a digital version with a pattern more like a series of steps than a wave form.